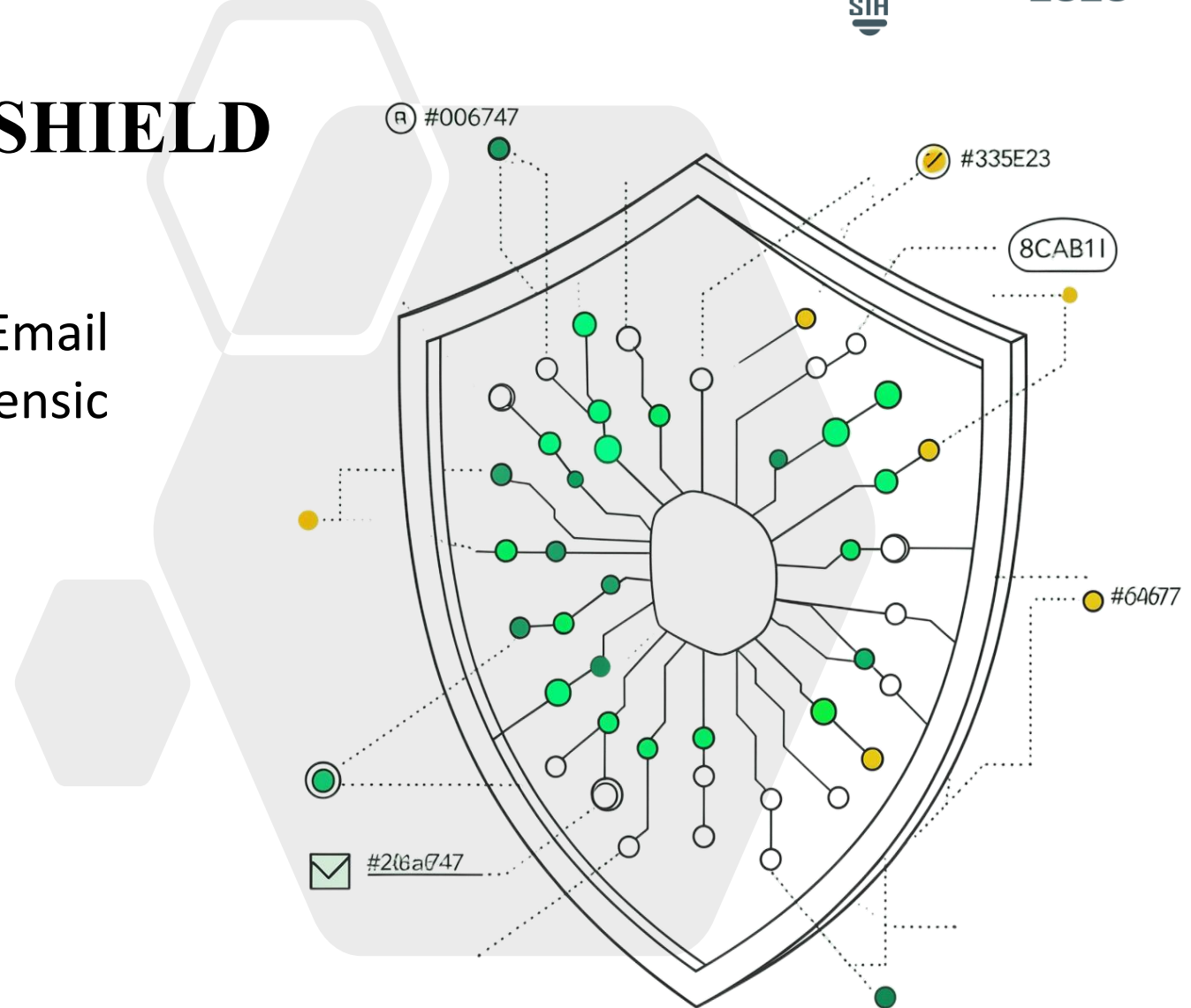


TRACE SHIELD

- **Problem Statement ID – SIH26106**
- **Problem Statement Title - AI-Powered Email Threat Detection, GeoLocation & Forensic Intelligence Platform**
- **PS Category- Software**
- **Team –**
 - 1) **ADITYA YADAV (TEAM LEADER)**
 - 2) **VAIBHAV PRAKASH**
 - 3) **MEISHA JAIN**
 - 4) **SHREYASH NIWATE**
 - 5) **ROYDEN MACHADO**
 - 6) **SAMARTH BADGUJAR**



THE MANDATE // PROBLEM SIH26106

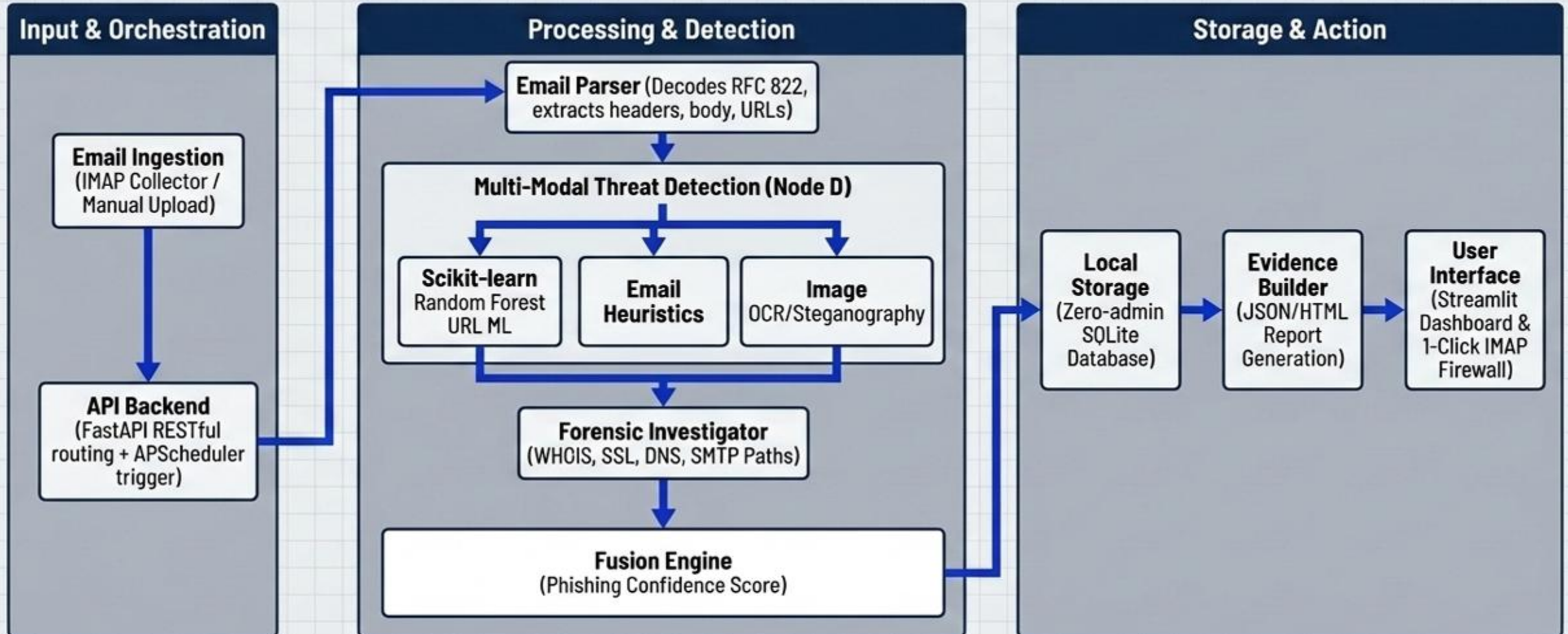
AI-Powered Email Threat Detection, GeoLocation, and Forensic Intelligence Platform

THE GAP: Current security tools act as **"Black Boxes."** They block malicious content but lack the deep forensic tracing required to map fraudulent origins.

- ✓ Reconstruct complete SMTP relay paths
- ✓ Analyze originating IPs & geolocation
- ✓ Validate SPF, DKIM, and DMARC alignment
- ✓ Generate explainable, confidence-based fraud risk assessments

CAPABILITY	LEGACY SECURE EMAIL GATEWAYS	PROPOSED API-NATIVE AI
Threat Detection	Static signature matching and known blocklists.	Multi-Modal AI (NLP Intent Parsing + Scikit-Learn Random Forest on 20 URL features).
Forensic Tracing	None (Black Box binary blocking).	Deep OSINT enrichment, Geographic IP hopping, and SMTP Timeline mapping (The Glass Box).
Deployment & Privacy	Cloud-dependent (High risk of classified data leakage).	Sovereign, Zero-Cloud Leakage (Local SQLite WAL storage with open-weight models).

TECHNICAL APPROACH



Technical Feasibility, Non-Disruptive Adoption, and Explicit Risk Mitigations (Addressing Hard Forensics Realities).

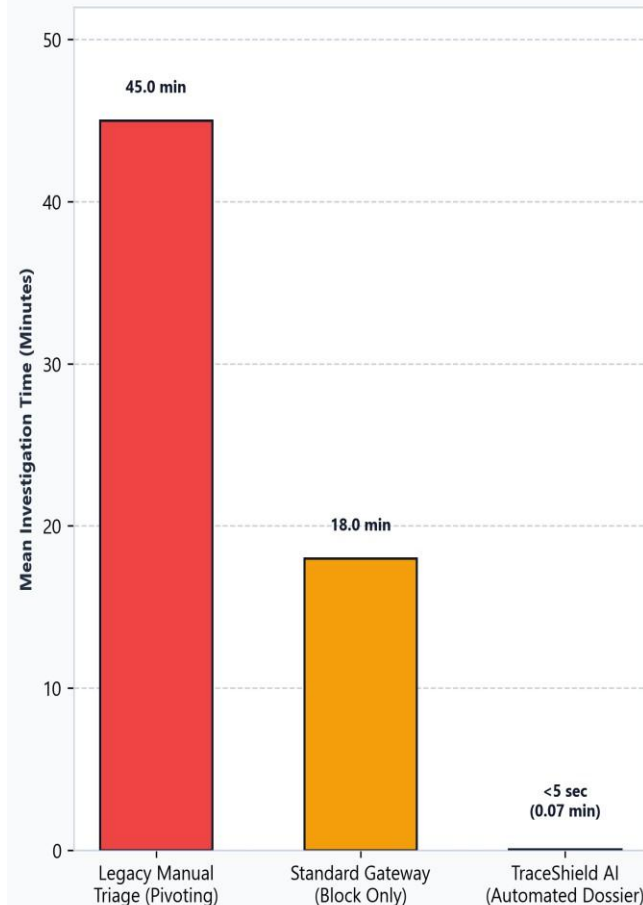
FEASIBILITY & OPERATIONAL VIABILITY

- **Technical Feasibility:** Built on standardized RFC 5322/7489 protocols; containerized microservices run lightweight on-premises or cloud.
- **Operational Viability:** Non-disruptive integration via API webhooks; ingests logs from existing gateways without altering mail MX routing.
- **Institutional Scalability:** Zero-learning-curve reporting button for staff/students; structured case view for Tier-1/2 SOC teams.

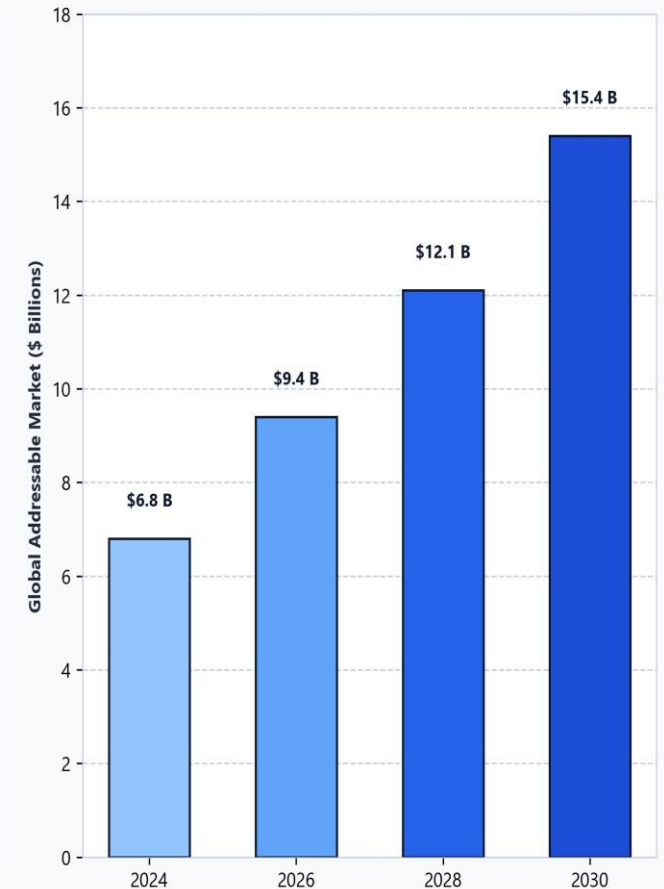
POTENTIAL RISKS & ENGINEERING MITIGATIONS

- **AI Hallucination / Drift:** Mitigated via deterministic rule engine baseline + inspectable reason codes + analyst confirmation override.
- **IP Geolocation Obfuscation:** VPN/TOR/Proxy context tagging with confidence intervals; explicitly prevents reckless human accusations.
- **Data Privacy & Compliance:** Local cryptographic hashing, strict RBAC, automated PII masking, and configurable data retention policies.

ANALYST TRIAGE TIME PER CASE
(Lower is Better — 98% Reduction)



GLOBAL EMAIL FORENSICS & SECURITY MARKET
(CAGR 14.6% driven by AI Phishing & BEC)



IMPACT AND BENEFITS



Delivering end-to-end resilience against zero-day phishing, BEC, and account takeovers for national and enterprise security.

Pre-Attack Visibility
(Dark Web Scope)



Zero-Day Exposure Mitigation

Continuously monitors underground markets and stealer logs. Neutralizes compromised credentials before they are weaponized into Account Takeovers (ATO).



Post-Attack Forensics
(Automated Response)

Rapid Attribution & Remediation

Extracts geolocation, domain intelligence, and forensic Indicators of Compromise (IOCs). Automates post-delivery inbox sweeping across the network in minutes, rather than hours.



Real-Time Defense
(Behavioral AI)

Synchronous Threat Neutralization

Applies Natural Language Processing (NLP) to detect Behavioural Baseline Deviation.

Blocks sophisticated Business Email Compromise (BEC) and VIP impersonation synchronously, without relying on known signatures.



Strategic Ecosystem
(National Impact)

Institutional Cyber Resilience

Prevents massive financial fraud, ensures secure multi-channel collaboration, and feeds correlated intelligence directly into existing SIEM/SOAR infrastructures.

YESTERDAY: Reactive Gateways. Relying on static signatures.
Blind to internal east-west traffic.

TODAY'S IMPACT: Proactive Intelligence. API-native integration.
Behavioral context. Explainable AI (XAI) for rapid SOC decision-making.

IETF Internet & Protocol Standards

- **RFC 5322**

Internet Message Format — authoritative structural standard for electronic messages

- **RFC 7208 (SPF)**

Sender Policy Framework — protocol authorizing sending SMTP client IPs

- **RFC 6376 (DKIM) & RFC 7489 (DMARC)**

Cryptographic domain-key validation and policy enforcement

Cybersecurity Frameworks & Directives

- **NIST SP 800-177 Rev 1**

Federal guidelines for trustworthy email and spoofing defense

- **MITRE ATT&CK Framework**

T1566 (Phishing), T1566.002 (Spearphishing), T1566.001 (Attachment)

- **CERT-In National Directives**

Mandatory cyber incident reporting and forensic evidence preservation guidelines

Federal & Global Threat Intelligence

- **Verizon DBIR (2024/2025)**

Phishing and BEC are root initial access vectors in >60% of enterprise breaches

- **FBI IC3 Internet Crime Report**

Business Email Compromise causes over \$2.9 Billion in annual global losses

- **Microsoft Digital Defense Report**

Analyzes adversary abuse of legitimate cloud relays and multi-stage evasion techniques

- **Use of LLM's, AI model's and Agentic AI**

For huge research purposes and organization of our project for smooth handling

Source	Detail
Open Source Repository	github.com/driveadityayadav18-art/sih26106-mvp
Golden Dataset Validation	444 automated tests against curated real-world attack vectors